

1. Executive Summary & Safety Invariant

SafeDig is an automated, AI-augmented safety validation platform engineered to prevent utility strikes during civil excavations in the United Kingdom. Under UK statutory frameworks (HSG47 and CDM 2015), striking buried infrastructure carries catastrophic risks: gas explosions, electric flashovers, hospital blackouts, and unlimited financial and criminal liabilities. SafeDig re-validates upstream utility claims against physical drawings.

NON-NEGOTIABLE DOMAIN INVARIANT (HSG47 COMPLIANCE)

SAFE_MODE is permanently active. Every detected hazard is surfaced to a certified human safety engineer. Zero escaped hazards is the non-negotiable target. Every ambiguous, corrupted, or contradictory document must fail toward HUMAN_REVIEW or BLOCKED, NEVER toward AUTO_CLEAR. A false negative is never traded for a cleaner false-positive statistic.

2. The 8 Chronological Pipeline Phases

Phase	Primary Code Modules	Operational Role & Scope
Phase 1: Ingestion & Paths	src/utlils/paths.py , src/index/parser.py	Self-healing path resolution; parses index.xlsx; separates...
Phase 2: PDF Structure	src/pdf/inspector.py , src/domain/document.py	PyMuPDF opens byte stream, computes SHA-256 hash, classifi...
Phase 3: Knowledge Base	src/legends/ , src/warnings/	Identifies statutory provider (UKPN, SGN, Cadent); loads w...
Phase 4: Spatial AOI	src/aoi/detector.py	Extracts red/magenta dashed site boundary; converts 0/90/1...
Phase 5: Dual CV Scan	src/detection/engine.py , src/vector/ , src/cv/	Channel A vector extraction (excludes dashed AOI) + Channe...
Phase 6: Spatial Reconcile	src/spatial/engine.py , src/reconciliation/	Applies 5m LV / 15m HP Gas buffers; computes 2D intersecti...
Phase 7: The 17 Gates	src/policy/gates.py , src/policy/engine.py	Simultaneously evaluates all 17 gates. Priority routing to...
Phase 8: LangGraph & Audit	src/orchestration/graph.py , safedig.db	Stateful DAG execution; Ollama Qwen 2.5 advisory summary; ...

Phase 1: Ingestion & Self-Healing Path Resolution

Self-Healing Path Translation

[src/utls/paths.py](#)

When repositories are cloned across different machines, index.xlsx and report metadata often store foreign machine paths (e.g. d:/Safedig_AG/data/...). `resolve_folder_path()` and `resolve_file_path()` dynamically inspect parent directories, resolving the folder basename with `os.path.basename`.
Impact: Solves 'Index file not found' across foreign workstations automatically.

Spreadsheet Parsing (index.xlsx)

[src/index/parser.py](#)

Reads index.xlsx via pandas and openpyxl. Extracts statutory disclosure fields: `utility_name`, `is_asset_present` (Status='Yes'/'No'), `raw_warning`, and `raw_comments`. Produces a typed `IndexRecord` list.
Impact: Validates that all utility rows are parsed without loss or truncation.

Deterministic Inventory Matching

[src/ingestion/inventory.py](#)

A utility pack contains dozens of PDFs (customer letters, hospital directions, valve leaflets). `inventory.py` tokenizes filenames, computes fuzzy string similarity against `utility_name`, and filters out excluded keywords.
Impact: Assigns `DocumentResolutionStatus`: UNIQUE (1-to-1 match), EXCLUDED, or AMBIGUOUS (triggers BLOCKED).

Phase 2 & 3: PDF Structure & Knowledge Resolution

PDF Stream Health & Modality

[src/pdf/inspector.py](#), [src/domain/document.py](#)

PyMuPDF opens the PDF file, computes a SHA-256 cryptographic byte hash, and inspects internal content stream operations. Classifies drawings as VECTOR (native path commands), RASTER (bitmap scans), or HYBRID.
Impact: Rejects corrupted or password-encrypted files early with BLOCKED.

Statutory Warning Catalogue

[src/domain/warning_catalogue.py](#)

Loads the statutory hazard taxonomy for the provider: voltage classifications (11kV, 33kV, 132kV), gas pressure bands (LP, MP, IP, HP > 7 bar), and required HSG47 safety controls (hand-dig, trial holes, clearance zones).
Impact: Establishes legal compliance baseline before spatial analysis.

Cartographic Legend Profile

[src/domain/legend.py](#)

Loads the provider's drawing legend profile: target RGB colors (e.g. UKPN 11kV = #FF0000, SGN LP Gas = #FFCC00), line stroke widths in points, and color tolerance bands (Delta E <= 28.0 Euclidean RGB distance).
Impact: Allows automated computer vision to discriminate between gas, electric, and water lines.

Phase 4: Spatial AOI Extraction & Rotation Mapping

The Area of Interest (AOI) represents the physical boundary of the planned civil excavation. SafeDig resolves three major cartographic challenges in `src/aoi/detector.py`:

- 1. Rotation Coordinate Conversion: PyMuPDF's `get_drawings()` returns coords in unrotated mediabox space. When drawings are rotated (0, 90, 180, 270 deg), `detector.py` translates visual page rects back to mediabox:
 - 90 deg: `x_un = y_vis, y_un = mh - x_vis`
 - 180 deg: `x_un = mw - x_vis, y_un = mh - y_vis`
 - 270 deg: `x_un = mw - y_vis, y_un = x_vis`
- 2. Dashed Perimeter Search: Detects dashed vector boundaries forming the dig boundary (red, magenta, yellow).
Perimeter Validation: Validate closed Shapely Polygon (`coords[0] == coords[-1]`, vertices ≥ 4 , area > 100).
~~Conservative Fallback: If no boundary line is found, uses the active map canvas so no assets are missed.~~

Channel A: PyMuPDF Vector Engine src/vector/analyzer.py Directly parses vector path commands (moveto, lineto, curveto). Yields zero pixel distortion. CRITICAL INVARIANT: Enforces <code>exclude_dashed=True</code> so that the dashed site boundary itself is never confused with an underground cable. Matches solid vectors against legend RGB and stroke width bands.	
Channel B: OpenCV 300 DPI Raster CV src/cv/ Renders high-resolution 300 DPI bitmaps to inspect legacy scanned maps that lack vector paths. Applies HSV color thresholding, morphological closing (<code>cv2.morphologyEx</code>), and contour hierarchy analysis to detect point symbols (substations, link boxes, valves, manholes, chambers).	
OCR / Text Label Booster src/pdf/extractor.py Extracts text blocks located inside the AOI bounding box. Correlates detected line vectors with technical abbreviations ('11kV', 'HV', 'HP GAS', 'SUBSTATION'). If a high-voltage tag is found near a line, severity is immediately promoted to CRITICAL.	

Phase 6: Spatial Buffer Math & Reconciliation Outcomes

Civil excavations cannot take place immediately adjacent to pressurized or high-voltage services.
SafeDig expands the excavation polygon using Shapely 2D buffer geometry:

- Standard Hand-Dig Zone: +5.0m buffer around all low/medium voltage cables and distribution pipes (HSG47).
- High-Pressure Gas / EHV Corridor: +15.0m statutory clearance zone around Cadent/SGN transmission pipelines.

Computes exact 2D geometric intersection: buffered_aoi.intersects(asset_geom) and aoi.distance(asset_geom).

CONFIRMED_CLEAN	Upstream: 'No Assets' SafeDig: Zero Intersecting Vectors	AUTO_CLEAR
MATCH (Low/Med)	Upstream: 'Assets Present' SafeDig: Detected Intersecting Lines	HUMAN_REVIEW
MATCH (High/Crit)	Upstream: 'Assets Present' SafeDig: Detected 11kV/HP Gas	HUMAN_REVIEW
MISSED_WARNING	Upstream: 'No Assets' SafeDig: Detected Buried Infrastructure	HUMAN_REVIEW
POSSIBLE_FALSE_POS	Upstream: 'Assets Present' SafeDig: Buffer Completely Clean	HUMAN_REVIEW

Phase 7: The 17 Mandatory Release Gates & Policy Ladder

Ladder Priority	Trigger Condition	Decision
Step 1: Clean Claim Early Exit	index_record.is_asset_present == False (Status='No')	AUTO_CLEAR
Step 2: Missing File Early Exit	document is None OR document.is_corrupted	BLOCKED
Step 3: Ambiguous Match Early Exit	index_record.resolution_status == AMBIGUOUS	BLOCKED
Step 4: Missed Warning Escalation	reconciliation.outcome == MISSED_WARNING	HUMAN_REVIEW
Step 5: False Positive Escalation	reconciliation.outcome == POSSIBLE_FALSE_POSITIVE	HUMAN_REVIEW
Step 6: High Hazard Escalation	outcome == MATCH and Gate 12 Fails (11kV / HP Gas)	HUMAN_REVIEW
Step 7: Geometry/Legend Escalation	Gate 07 (Legend) or Gate 08 (AOI) Failed	HUMAN_REVIEW
Step 8: Final Gate Sweep	ALL 17 Gates passed and evidence package complete	AUTO_CLEAR

Phase 8: LangGraph Workflow, Advisory LLM & Forensic Audit

LangGraph State Machine (src/orchestration/graph.py):

- Orchestrates nodes: ingest_and_index ... process_qa_and_policy ... llm_advisory ... finalize ... persist_to_db.
- Routes jobs requiring human review through llm_advisory_node; clean jobs proceed straight to finalize_report.

Advisory LLM & Zero Validation Authority:

- Queries local Ollama runtime (Qwen 2.5 7B / Llama 3.2 3B) to generate plain-English operator notes.
- INVARIANT: The LLM has zero authority to override, alter, or auto-clear any safety gate.
- Oms offline fallback explainer operates with zero latency if Ollama is unavailable.

Forensic Audit & SQLite Persistence (safedig.db):

Interactive Web Console & QA Workspace (<http://127.0.0.1:8000>)

- Computes SHA-256 digital signatures of original PDF, index row, detection GeoJSON, and timestamps.
- Persists immutable audit records via SQLAlchemy for legally defensible compliance under HSE inquiries.

Animated Radar Scanner Loader	src/api/static/index.html & app.js
When an engineer inspects a job, #ws-map-loader displays an animated radar scanner with spinning gradient rings while high-resolution map tiles load. Automatically fades out when the DOM image finishes rendering.	
Hardware-Accelerated Pan & Zoom	Canvas / DOM Viewport
Supports fluid drag-and-pan and zooming from 50% to 600%. Features instant AOI centering to immediately focus on the excavation boundary and detected hazards.	
Live 17 Gates Status Grid	Interactive Policy Matrix
Renders the 17 gates dynamically with green checkmarks or red failure badges. Clicking any gate highlights the relevant geometry or displays the exact rejection rationale.	
Human Override Sign-Off Audit	src/api/routes/qa.py
Certified safety engineers can record an override (Accept Risk & Authorize or Reject Permit). The engineer's credential, action, and timestamp are permanently written to safedig.db.	